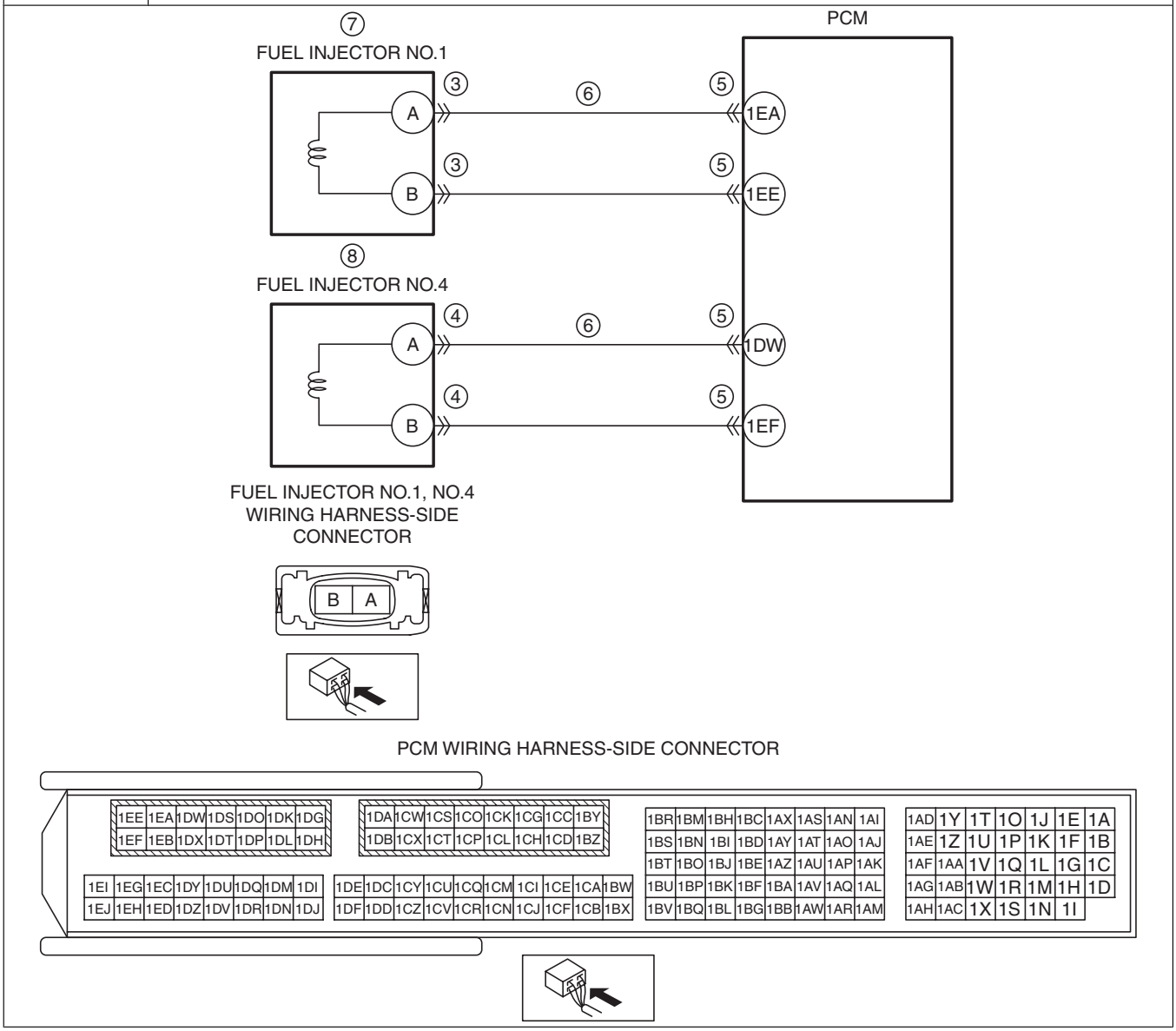
DTC P2146:00 [SKYACTIV-D 1.5]

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DTC P2146:00	Fuel injector No.1 and No.4 circuit abnormal operation
DETECTION CONDITION	• When the following condition is met, the PCM detects the control current at fuel injectors No.1 and No.4 as **less than the 0.67 A** value for continuous **8 times**. **MONITORING CONDITIONS** — During engine running — Battery voltage: **8 V or more** — **2 s or more** have elapsed since PCM launch — During fuel injection control — The following DTCs are not detected: • PCM internal processor malfunction: P0606:00 • Internal PCM malfunction: P062B:00 **Diagnostic support note** • This is an intermittent monitor (fuel system). • The check engine light illuminates if the PCM detects the above malfunction condition during the first drive cycle. • FREEZE FRAME DATA/Snapshot data is available. • DTC is stored in the PCM memory.
FAIL-SAFE FUNCTION	• PCM restricts engine torque. • Inhibits the EGR control. • Inhibits the auto DPF regeneration control. • Inhibits the compulsory DPF regeneration control. • Inhibits engine-stop by operating the i-stop function. • PCM restricts engine-transaxle integration control. (ATX)
POSSIBLE CAUSE	• Fuel injector No.1 connector or terminals malfunction • Fuel injector No.4 connector or terminals malfunction • PCM connector or terminals malfunction • Open circuit in wiring harness between the following terminals: — Fuel injector No.1 terminal A—PCM terminal 1EA — Fuel injector No.1 terminal B—PCM terminal 1EE — Fuel injector No.4 terminal A—PCM terminal 1DW — Fuel injector No.4 terminal B—PCM terminal 1EF • Fuel injector No.1 malfunction • Fuel injector No.4 malfunction • PCM malfunction

DTC P2146:00	Fuel injector No.1 and No.4 circuit abnormal operation



Diagnostic Procedure

STEP	INSPECTION		ACTION
1	**RECORD VEHICLE STATUS AT TIME OF DTC DETECTION TO UTILIZE WITH REPEATABILITY VERIFICATION** **Note** • Recording can be facilitated using the screen capture function of the PC. • Record the FREEZE FRAME DATA/snapshot data on the repair order.	—	Go to the next step.
2	**VERIFY RELATED SERVICE INFORMATION AVAILABILITY** • Verify related Service Information availability. • Is any related Service Information available?	Yes	Perform repair or diagnosis according to the available Service Information. • If the vehicle is not repaired, go to the next step.
		No	Go to the next step.

STEP	INSPECTION		ACTION
3	**INSPECT FUEL INJECTOR NO.1 CONNECTOR CONDITION** • Switch the ignition off. • Disconnect the fuel injector No.1 connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 9.
		No	Go to the next step.
4	**INSPECT FUEL INJECTOR NO.4 CONNECTOR CONDITION** • Disconnect the fuel injector No.4 connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 9.
		No	Go to the next step.
5	**INSPECT PCM CONNECTOR CONDITION** • Disconnect the PCM connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 9.
		No	Go to the next step.
6	**INSPECT FUEL INJECTOR CIRCUIT FOR OPEN CIRCUIT** • Verify that the fuel injector No.1, fuel injector No. 4 and PCM connectors are disconnected. • Inspect for continuity between the following terminals (wiring harness-side): — Fuel injector No.1 terminal A—PCM terminal 1EA — Fuel injector No.1 terminal B—PCM terminal 1EE — Fuel injector No.4 terminal A—PCM terminal 1DW — Fuel injector No.4 terminal B—PCM terminal 1EF • Is there continuity?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between the following terminals: • Fuel injector No.1 terminal A—PCM terminal 1EA • Fuel injector No.1 terminal B—PCM terminal 1EE • Fuel injector No.4 terminal A—PCM terminal 1DW • Fuel injector No.4 terminal B—PCM terminal 1EF **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to Step 9.
7	**INSPECT FUEL INJECTOR NO.1** • Inspect the fuel injector No.1. (See FUEL INJECTOR INSPECTION [SKYACTIV-D 1.5].) • Is there any malfunction?	Yes	Replace the fuel injector No.1, then go to Step 9. (See FUEL INJECTOR REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
		No	Go to the next step.
8	**INSPECT FUEL INJECTOR NO.4** • Inspect the fuel injector No.4. (See FUEL INJECTOR INSPECTION [SKYACTIV-D 1.5].) • Is there any malfunction?	Yes	Replace the fuel injector No.4, then go to the next step. (See FUEL INJECTOR REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
		No	Go to the next step.
9	**VERIFY DTC TROUBLESHOOTING COMPLETED** • Always reconnect all disconnected connectors. • Clear the DTC from the PCM memory using the M-MDS. (See AFTER REPAIR PROCEDURE [SKYACTIV-D 1.5].) • Perform the KOER self test. (See KOEO/KOER SELF TEST [SKYACTIV-D 1.5].) • Is the same DTC present?	Yes	Repeat the inspection from Step 1. • If the malfunction recurs, replace the PCM. (See PCM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) Go to the next step.
		No	Go to the next step.
10	**VERIFY AFTER REPAIR PROCEDURE** • Perform the “AFTER REPAIR PROCEDURE”. (See AFTER REPAIR PROCEDURE [SKYACTIV-D 1.5].) • Are any DTCs present?	Yes	Go to the applicable DTC inspection. (See DTC TABLE [SKYACTIV-D 1.5].)
		No	DTC troubleshooting completed.